

The empirical recurrence for OEIS A298059: recovering a conjecture that is not written down

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Abstract

OEIS A298059 records “Empirical recurrence of order 71”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 72$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 71, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 71 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A298059 is “Number of $n \times 4$ 0..1 arrays with every element equal to 1, 3, 4 or 6 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 13, 2, 11, 32, 42, 104, 269, ...

The entry states:

Empirical recurrence of order 71 (see link above)

As of the “Last modified” line on the live entry (Jan 11 2018 08:00:10, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 72$ of the 256, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 144$ exact terms of the model returns order 71 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{71} c_i a(n-i),$$

c_1	2	c_{19}	2880322	c_{37}	15309076	c_{55}	−129626
c_2	3	c_{20}	3037114	c_{38}	2270453	c_{56}	−78001
c_3	43	c_{21}	1168516	c_{39}	−12291560	c_{57}	−43765
c_4	−92	c_{22}	−7769826	c_{40}	−6152440	c_{58}	28747
c_5	−137	c_{23}	−7756899	c_{41}	2835880	c_{59}	−3244
c_6	−766	c_{24}	−1075213	c_{42}	5681329	c_{60}	12392
c_7	1840	c_{25}	15751311	c_{43}	1350500	c_{61}	5955
c_8	2532	c_{26}	14625731	c_{44}	−3186188	c_{62}	4132
c_9	7659	c_{27}	−1270302	c_{45}	−1412414	c_{63}	−5736
c_{10}	−21039	c_{28}	−24233380	c_{46}	−178364	c_{64}	−4112
c_{11}	−26644	c_{29}	−20081611	c_{47}	1826896	c_{65}	−1252
c_{12}	−48668	c_{30}	6784503	c_{48}	392177	c_{66}	752
c_{13}	155093	c_{31}	28287505	c_{49}	−71838	c_{67}	560
c_{14}	182839	c_{32}	19295059	c_{50}	−805474	c_{68}	336
c_{15}	208399	c_{33}	−13301605	c_{51}	−273237	c_{69}	448
c_{16}	−788376	c_{34}	−24640651	c_{52}	210327	c_{70}	320
c_{17}	−875840	c_{35}	−11576083	c_{53}	324080	c_{71}	128
c_{18}	−610755	c_{36}	15972092	c_{54}	107576		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 71, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $71 + 4$ terms removed returns order 71 as well, so $\deg q_{\min} = 71 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 29 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $71 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 71 that this sequence satisfies — and that family turns out to have exactly one member.

References

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